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Course: BCSNT 6063 – Object Oriented Programming (2nd Semester)
Assignment Type: Practical / Tutorial
Submission: Screenshots + Program Files
Deadline: 22nd June 2026 (Monday)

Handwritten Codes :

1

```
public class Ques1Main{
        public class Student{
                private String name;
                private int rollNum;
                private String faculty;

                public Student(String name,int no, String faculty){
                        this.name= name;
                        this.rollNum= no;
                        this.faculty= faculty;
                }
                public String getName(){ return name;}
                public int getRollNum(){return rollNum;}
                public String getFaculty(){return faculty;}
              public void setName(String n){this.name=n;}
              public void setRollNum(int n){ this.rollNum=n;}
              public void setFaculty(String f){this.faculty=f;}
              public void display(){
                      System.out.println("<----Stud Prof ---->");
                      system.out.println("Student. Name:" + name);
                      System.out.println("Stud RollNumber:" + rollNum);
                      System.out.println("Stud Faculty:" + faculty);
              }
        }
        public static void main(String[] args){
                Student s1= new Student("Pratyush",20, "BCS");
                Student s2 =new Student("Nishan", 16, "BCS");

                s1.display();
                s2.display();
                s1.setName("Ruben");
                s1.display();
        }
}
```

②

```
public class Ques2Main{
    public class Employee{
        protected int id;
        protected String name;
        public Employee(int id, String n){
                this.id = id;
                this.name = n;
        }
        public void display(){
            System.out.println("<-Employees->");
            System.out.println("Emp id :" + id);
            System.out.println("Emp name:" + name);
        }.
    }
    public class Teacher extends Employee{
            protected String sub;
            public teacher(int id, String name, String sub){
                    super(id,name);
                    this. subg = sub;
            } @Override
            public void display(){
                    System.out.println(" Teacher ");
                    System.out.println(" ID :" + id);
                    System.out.println("Name:" + name);
                    System.out.println("Subject:" + sub);
            }
    }
    public class AdminStaff extends Employee{
            private final String department;
            public AdminStaff(int id, String n, String dep){
                    super(id,n);
                    this.department= dep;
            }
```

```
    @Override
    public void display(){
            System.out.println("AdminStaff");
            System.out.println("ID:"+id);
            System.out.println("Name:"+n);
            System.out.println("Department:"+dep);
    }
    public static void main(String[] args){
        Employee e = new Employee(1, "Alan");
        AdminStaff a = new AdminStaff(2, "John", "BCS");
        Teacher t = new Teacher(3, "Harry", "Magic");

        e.display();
        a.display();
        t.display();
    }
}
```

(3)

```
public class Ques3Main{
  public class Course{
        private int id;
        private String n;
        public Course(int id, String n){
            this.id=id;
            this.n = n;
        }
        public int getID(){ return this.id;}
        public String getName(){ return this.n;}
        @Override
        public String toString(){
            return "Course id:"+this.~~name~~ id +"-"+ "Name:"+this.n;
        }
}
```

```
public class student{
    int num;
    String n;
    Arraylist<Course> c;
    Arraylist<Course> rc;
    public Student{int num, String n}{
        this.num = num;
        this.n = n;
        this.c = new Arraylist<>();
        this.rc = new Arraylist<>();
    }
    public void RegCourse(Course c){
        .rc.add(c);
    }
    public void display(){
        System.out.println("\n Std Detail ");
        System.out.println(" Name :" + n);
        System.out.println(" Roll Num:" + num);
        System.out.println(" Reg Course:");
        for (Course c: rc){
            System.out.println(c);
        }
    }
}
public static void main(String[] args){
        Course[] cs= { new Courgse(101,"CS"),new Course(102,"OOP")};
        Student s1=new Student(1,"Pratyush");
        s1.rc(cs[0]);
        s2.rc(cs[1]);
        s1.display();
    }
}
```

(4)

```
public class Ques4Main{
    abstract class Std{
        String n;
        int id;
        Std(String n, int id){
            this.n = n;
            this.id = id;
        }
        abstract double calc();
        public void display(){
            System.out.println(" Name:" + n);
            System.out.println(" ID:" + id);
        }
    }
    class UG extends Std{
        double credHr;
        double feeperCred;
        UG(String n, int id, double credHr, double feeperCred){
            super(n, id);
            this.credHr = credHr;
            this.feeperCred = feepercred;
        }
        @Override
        double calc(){
            return credHr * feeperCred;
        }
    }
    class G extends Std{
        double ffee;
        double rfee;
        G(String n, int id, double ffee, double rfee){
            super(n, id)
            this.ffee = ffee;
            this.rfee = rfee;
```

```
@Override
    double calc(){
        return ffee+rfee;
    }
}
public static void main(String[] args){
        Std u= new UG("Pratyush", 106, 20, 150);
        Std p= new G("Nishan", 202, 3000, 1200);
        u.display();
        System.out.println("UG Fee:"+u.calc());
        p.display();
        system.out.println("PG Fee:"+ p.calc());
    }
}
```

(5)

```
public class Ques5Main{
    public interface RP{
        double CG(double m);
        String GS(double m);
    }
    public class ED implements RP{
        @Override
        public double CG(double m){
            return m * 0.9;
        }
        @Override
        public String GS(double m){
            double g= CG(m);
            if (g>=85) return 'Distinction';
            else if (g>=70) return '1st class';
            else if (g>=50) return 'pass';
            else return 'Fail';}
}
```

```
public class MD implements RP{
        @Override
        public double CG(double m){
            return m;
        }
        public String GS(double m){
            double g= CG(m);
            if (g>=80) return 'Distinction';
          else if (g>=65) return '1st class';
          else if (g>=50) return 'pass';
            else return 'fail';
        }
    }
    public static void main(String[] args){
        RP e= new ED();
        RP m= new MD();

        double m= 75;
        system.out.println("Engineering:"+e.GS(m));
        System.out.println(" Management:"+m.GS(m));
    }   }
}
```

⑥

```
public class Ques6Main{
    public class Member{
        protected string n;
        public Member(string n){
            this.n =n;
        }
        public int borrow(){
            return 2;
        }
```

```
public void display(){
        System.out.println("\nName:"+n);
        System.out.println("BorrowLimit:"+ borrow());
    }
}
public class Teach extends Member{
    public Teach(string n){
        super(n);
    }
    @Override
    public int borrow(){
        return 10;
    }
}
public class Std extends Member{
    public std(string n){
        super(n);
    }
    @Override
    public int borrow(){
        return 4;
    }
}
public static void main(String[] args){
    Member S1= new std("Ram");
    Member t1 = new teach("Hari");

    s1.borrow();
    t1.borrow();
    s1.display();
    t1.display();
}
}
```

(7)

```
public class Ques7Main{
        public class Attendance{
            protected String n;
            protected int total;
            protected int attend;

            public Attendance(String n, int t, int a){
                this.n = n;
                this.total= t;
                this.attend= a;
            }

        public double calc(){
                if (attendance ==0) return 0;
                else { return ((double) attend/ total) * 100; }
        }
    }
    class Engineering extends Attendance{
        public Engineering(String n, int t, int a){
            super(n, t, a);
        }
        @Override
        public double calc(){
            double p= super.calc();
            return Math.floor(p);
        }
    }
public class Medical extends Attendance{
        public Medical(String n, int t, int a){
            super(n,t,a);
        }
```

```
    @Override
    public double calc(){
        double p= super.calc();
        return Math.ceil(p);
    }
}
public static void main(String[] args){
    Attendance e=new Engineering("Pratyush", 60 ,52);
    Attendance m=new Medical("Supreem", 60, 60);

    System.out.println("Eng Attendance : " + e.calc() + "%");
    System.out.println("Med Attendance : " + m.calc() + "%");
}
}
```

(8)

```
public class Ques8Main {
    public interface Noti{
        void sendNoti(String m, String r);
        void notiType();
    }
    public class Email implements Noti{
        @Override
        public void SendNoti(String m, String r){
            system.out.println("Sending email to:"+r);
            System.out.println(" Message :" +m);
        }
        @Override
        public void notiType(){
            System.out.println("It's email notification....");
        }
}
```

```
public class SMS implements Noti{
        @Override
        public void sendNoti(String m, string r){
                System.out.println("Sending SMS to " + r);
                System.out.println("Message:" + m);
        }
        @Override
        public void notiType(){
                System.out.println("It's SMS notification....");
        }
    }
  public static void main(String[] args){
        Noti e= new Email();
        e.sendNoti("Hello", "Pratyush");
        e.notiType();

        Noti s=new SMS();
        s.sendNoti("Hello", "Nishan");
        s.notiType();
    }
}
```

(9)

```
public class Ques9Main{
    public class Std{
        public static int counter = 1000;
        private int id;
        private String n;
        std(string n){
            this.n= n;
            this.id=counter++;
        }
```

```
    public void display(){
        System.out.printf("\n ID: %d  Name: %s ", this.id, this.n);
    }
}
public  static void main(String[] args){
    Std s1 = new Std("Pratyush");
    Std s2 = new  Std ("Taylah");
    s1.display();
    s2.display();
  }
}
```

(10)

```
public class Ques10Main{
    abstract class Scholarship{
        protected String n;
        public Scholarship(String n){
            this.n = n;
        }
        public abstract boolean isEligible();

        public void display(){
            System.out.println();
            if(isEligible()){
                System.out.println(name+ "is Eligible");
            }else{
                System.out.println(name + "isn't Eligible");
            }
    }
```

```
public class Merit extends Scholarship{
        private final double gpa;
        public Merit(String n,double gpa){
                super(n);
                this.gpa=gpa;
        }
        @Override
        public boolean isEligible(){
                return gpa>=3.6;
        }
}
public class Need extends Scholarship{
        private stat double in;
        public schNeed(string n, double in){
                super(n);
                this.in=in;
        }
        @Override
        public boolean isEligible(){
                return in<=20000;
        }
}
public static void main(String[] args){
        Scholarship s1 = new Merit("Trishna", 3.8);
        Scholarship s2= new Need ("Pratyush", 5000);

        s1.display();
        s2. display();
    }
}
```

Code Outputs :

1)

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaa

<---- Student Profile ----->
Student Name : Pratyush Tamrakar
Student RollNumber : 20
Student Faculty : BCS

<---- Student Profile ----->
Student Name : Nishan Subedi
Student RollNumber : 18
Student Faculty : BCS

<---- Student Profile ----->
Student Name : Mohit Oli
Student RollNumber : 20
Student Faculty : BCS

Process finished with exit code 0
```

2)

```
"C:\Program Files\Java\jdk-26.0.1\bin

<---- Employees ---->
Emp id : 1
Emp name : alan

<---- AdminStaff ----->
ID : 2
Name : John Wick
Department : BCS

<----- Teacher ----->
ID : 3
Name : Harry Potter
Subject : Magic

Process finished with exit code 0
```

3)

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaa

Available Courses
---------------------
Course ID : 101 Course Name : OOP
Course ID : 102 Course Name : Computer System
Course ID : 103 Course Name : Computer Architecture
Course ID : 104 Course Name : Networking
Course ID : 105 Course Name : Computer Ethics
---------------------------------------------
Registered sucessfully ....
Registered sucessfully ....


<---- Student Details ---->
Name : Pratyush Tamrakar
Roll Number : 18
Registerd Courses :
Course ID : 102 Course Name : Computer System
Course ID : 101 Course Name : OOP

Process finished with exit code 0
```

4)

```
"C:\Program Files\Java\jdk-26.0.1\bin\
Name : Pratyush
ID : 101
Undergraduate Fee: 3000.0

Name : Trishna
ID : 201
Graduate Fee: 4200.0

Process finished with exit code 0
```

5)

```
"C:\Program Files\Java\jdk-26.0.1\
Engineering: Pass
Management: First Class

Process finished with exit code 0
```

6)

```
"C:\Program Files\Java\jdk-26.0.1

Name : Ram
Burrow Limit : 4

Name : Shyam
Burrow Limit : 10

Process finished with exit code 0
```

7)

```
"C:\Program Files\Java\jdk-26.0.1\b
Engineering Attendance: 86.0%
Medical Attendance: 100.0%

Process finished with exit code 0
```

8)

```
Sending EMAIL to user@example.com
Message: Your order has been placed
It's a email notification ....

Sending SMS to +977-98XXXXXXXX
Message: Your OTP is 123456
It's a SMS notification ....

Process finished with exit code 0
```

9)

```
"C:\Program Files\Java\jdk-26.0.1\bin

ID: 1000   Name: Pratyush Tamrakar
ID: 1001   Name: Dipan Koirala
ID: 1002   Name: Mikesh Buddhathoki
ID: 1003   Name: Sushant Shrestha

Process finished with exit code 0
```

10)

```
Pratyush, you are eligible for Scholarship .

Dipan, you are not eligible for this scholarship.

Process finished with exit code 0
```

<-Thank You ->